

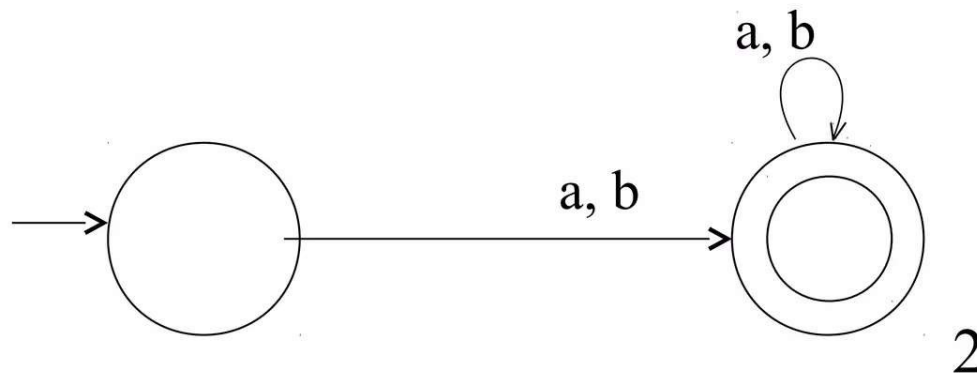
Recap Lecture 4



Regular expression of **EVEN-EVEN** language,
Difference between $a^* + b^*$ and $(a+b)^*$,
Equivalent regular expressions; sum, product
and closure of regular expressions; regular
languages, finite languages are regular,
introduction to finite automaton, definition of FA,
transition table, transition diagram.

Note

It may be noted that to indicate the initial state, an arrow head can also be placed before that state and that the final state with double circle, as shown below. It is also to be noted that while expressing an FA by its transition diagram, the labels of states are not necessary.



Example



$\Sigma = \{a,b\}$

States: x, y , where x is both initial and final state.

Transitions:

1. At state x reading a or b go to state y .
2. At state y reading a or b go to state x .

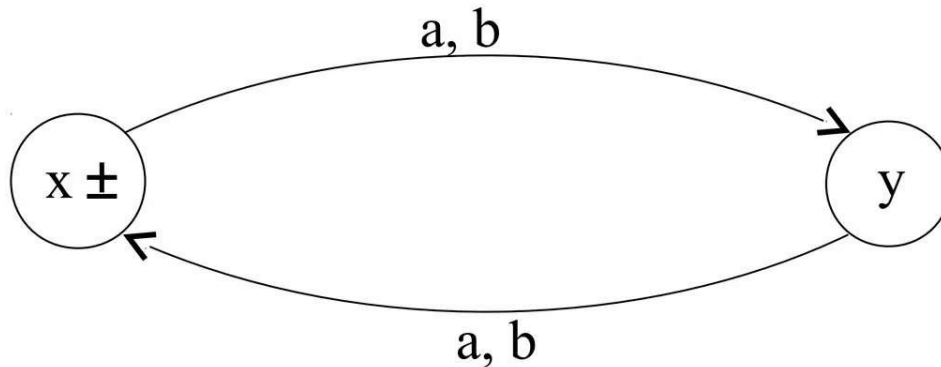
Example Continued ...

These transitions can be expressed by the following transition table

Old States	New States	
	Reading a	Reading b
$x \pm$	y	y
y	x	x

Example Continued ...

It may be noted that the previous transition table may be depicted by the following transition diagram.



Example Continued ...

The previous transition diagram is an FA accepting the language of strings, defined over $\Sigma=\{a, b\}$ of **even length**. It may be noted that this language may be expressed by the regular expression

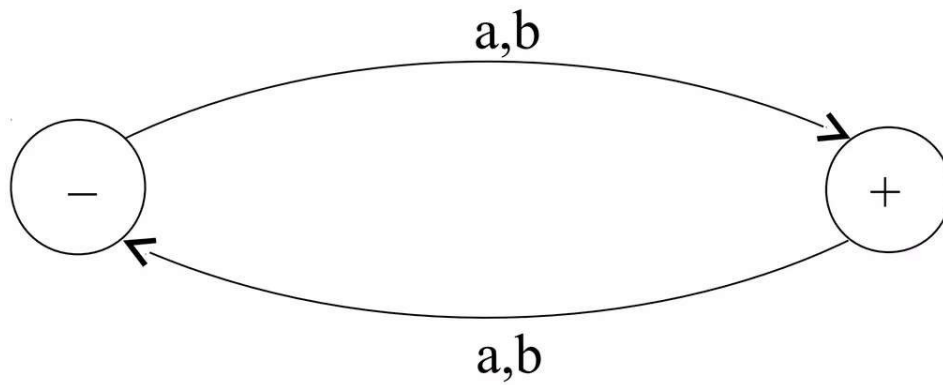
$$((a+ b) (a + b))^*$$

TASK

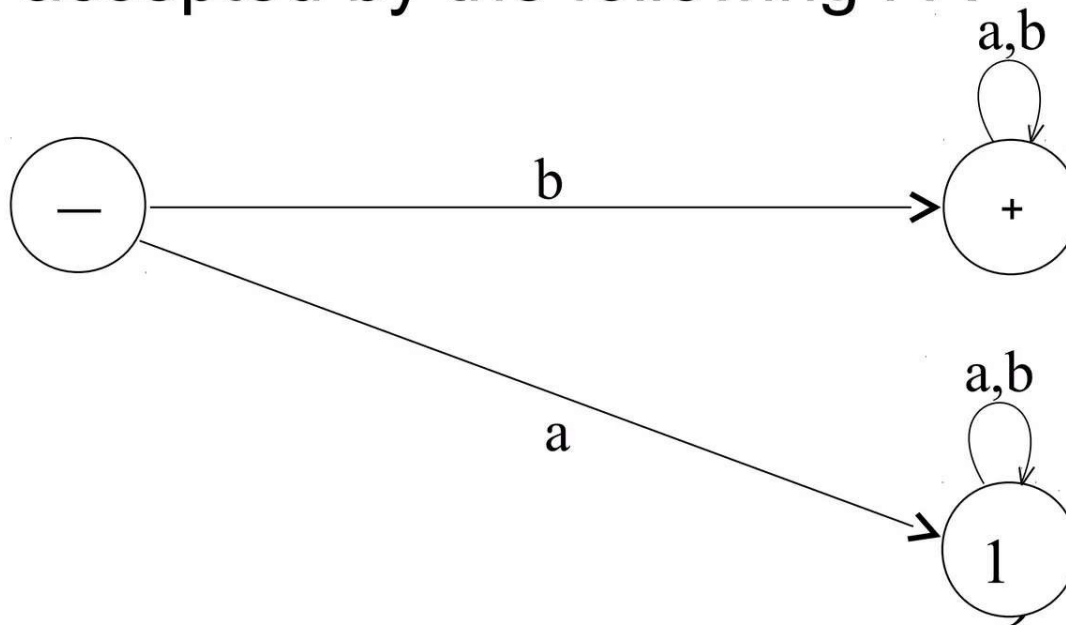


Build an FA for the language L of strings, defined over $\Sigma=\{a, b\}$, of odd length.

Solution of Task



Example: Consider the language L of strings, defined over $\Sigma=\{a, b\}$, **starting with b** . The language L may be expressed by RE $b(a + b)^*$, may be accepted by the following FA





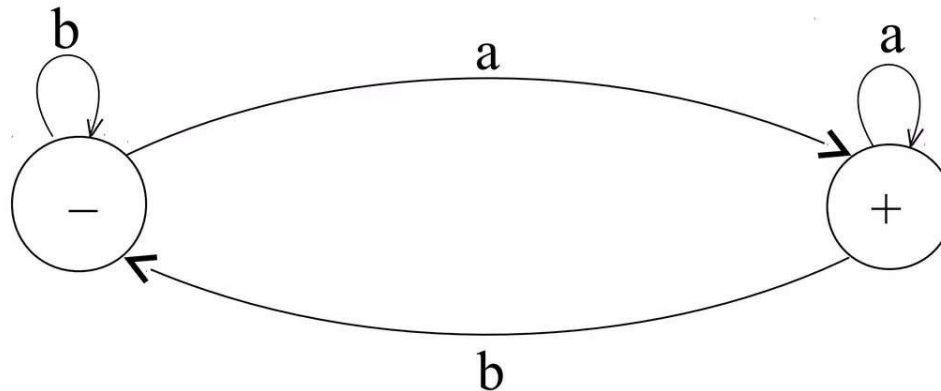
Example:

Consider the language L of strings, defined over $\Sigma=\{a, b\}$, **ending in a**. The language L may be expressed by RE

$$(a+b)^*a$$

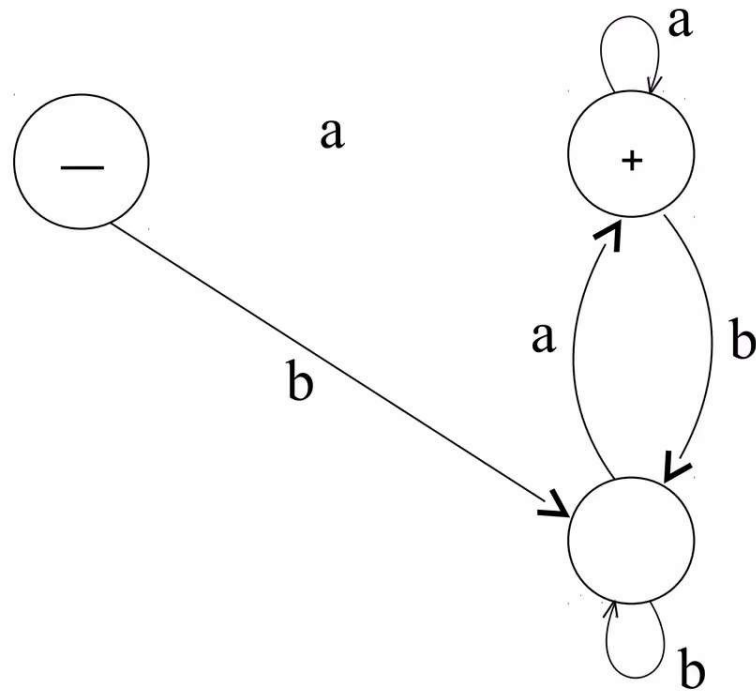
This language may be accepted by the following FA

Example Continued ...



There may be another FA corresponding to the given language.

Example continued ...



Note



It may be noted that corresponding to a given language there may be more than one FA accepting that language, but for a given FA there is a unique language accepted by that FA.

Note



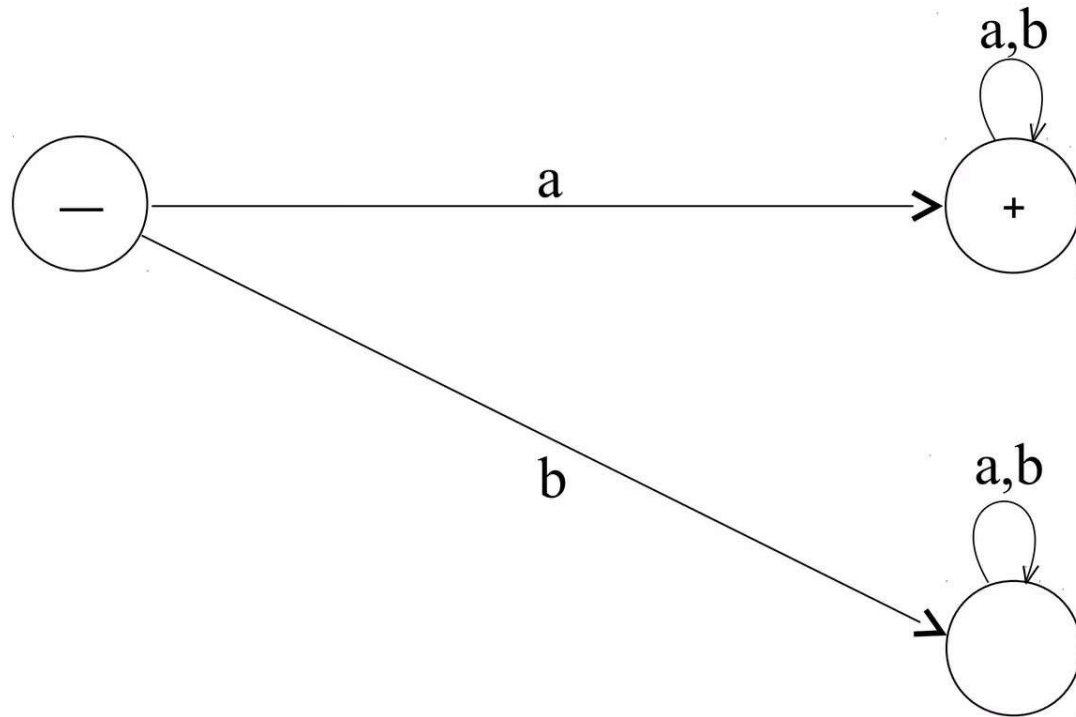
It is to be noted that given the languages L_1 and L_2 , where

L_1 = The language of strings, defined over $\Sigma = \{a, b\}$, **beginning with a**

L_2 = The language of strings, defined over $\Sigma = \{a, b\}$, **not beginning with b**

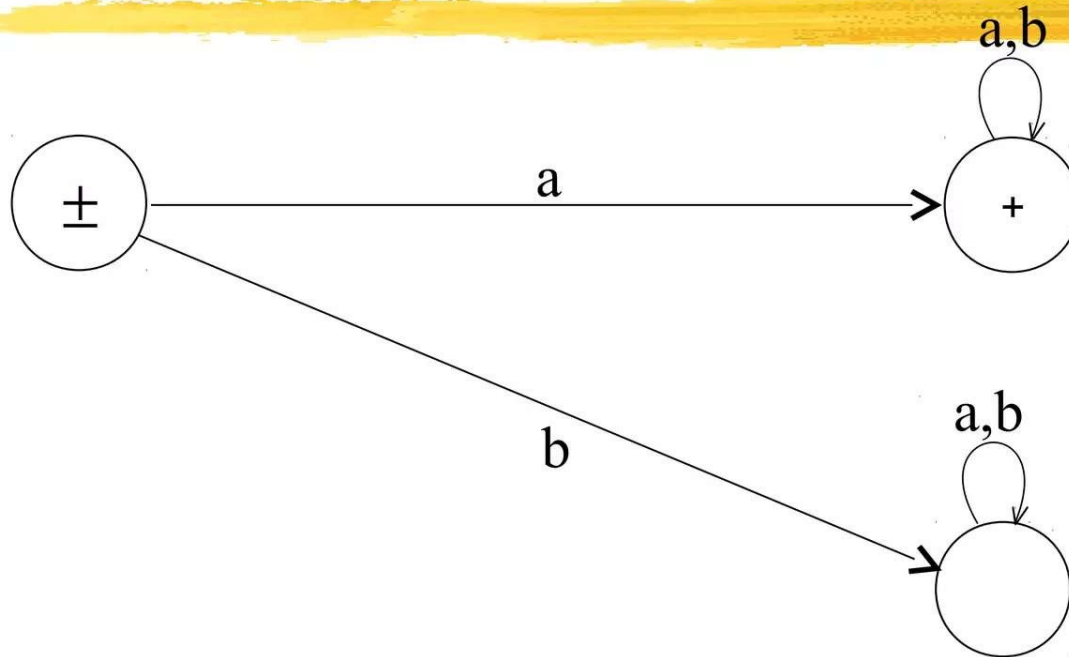
The Λ does not belong to L_1 while it does belong to L_2 . This fact may be depicted by the corresponding transition diagrams of L_1 and L_2 .

FA₁ Corresponding to L₁



The language L_1 may be expressed by the regular expression $a(a + b)^*$

FA₂ Corresponding to L₂



The language L_2 may be expressed by the regular expression $a(a + b)^* + \Lambda$

Example

Consider the Language L of Strings of **length two or more**, defined over $\Sigma = \{a, b\}$, **beginning with and ending in same letters.**

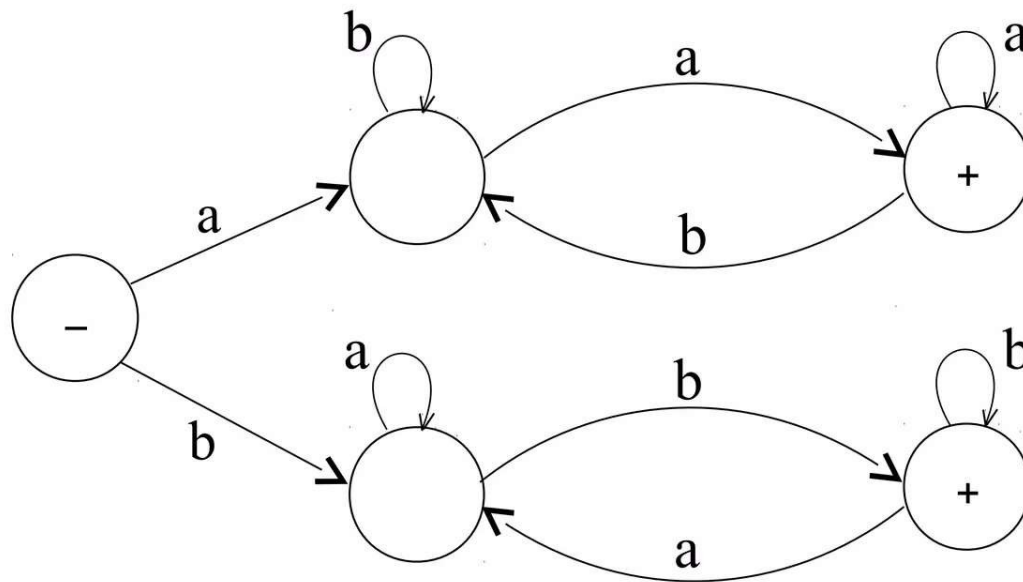
The language L may be expressed by the following regular expression

$$a(a + b)^* a + b(a + b)^* b$$

It is to be noted that if the condition on the length of string is not imposed in the above language then **the strings a and b will then belong to the language.**

This language L may be accepted by the following FA

Example Continued ...



Task



Build an FA accepting the Language L of Strings, defined over $\Sigma = \{a, b\}$, **beginning with and ending in same letters.**

Summing Up



Different notations of transition diagrams,
languages of strings of **even length**, **Odd length**, **starting with b**, **ending in a**,
beginning with b, **not beginning with b**,
beginning and ending in same letters;